Abhishek Surendra Shamkuwar

M.Tech., Signal Processing and Machine Learning | Dr. B. R. Ambedkar NIT Jalandhar, India
abhishekshamkuwar19@gmail.com | GitHub | LinkedIn | IEEE Xplore

RESEARCH INTERESTS

I'm interested in using machine learning, computer vision, and signal processing to build AI that is actually useful to people. Healthcare is where most of my work has been so far. For my thesis, I built a model that predicts sepsis early from ICU data, which meant combining different kinds of patient data, like vital signs and lab results, in a single model. Next, I want to work with brain signals (EEG) and medical images such as CT and MRI. I'm also drawn to human–computer interaction and robotics, and I'm curious how the same ideas could be used there, for example in brain-computer interfaces or robots that respond to people. Wherever I end up, I'd like to learn how to make these models accurate and something people can trust.

EDUCATION

M.Tech. in Signal Processing and Machine Learning, Dr. B. R. Ambedkar National Institute of Technology Jalandhar, Punjab, India — 2024 – 2026
CGPA: 8.38/10. Thesis: *Early Sepsis Prediction Using Multimodal Deep Learning*. Supervisor: Dr. Manjeet Singh.
Coursework: Machine Learning, Applied Machine Learning, Deep Learning, Statistical Signal Processing, Advanced Signal Processing, Probability and Stochastic Processes, Applied Linear Algebra and Optimization.

B.Tech. in Electronics and Communication Engineering, Indian Institute of Information Technology (IIIT) Sri City, Chittoor, Andhra Pradesh, India — 2020 – 2024
Coursework: Pattern Recognition, Digital Signal Processing, Digital Image Processing, Data Structures and Algorithms, Object-Oriented Programming, Embedded Systems, Control Systems.

PUBLICATIONS

Journal

[J1] A. Kar, P. K. Shill, **A. S. Shamkuwar**, S. Saini, Vasundhara, S. Shoba, and M. G. Christensen, "Adaptive Improved Primal-Dual Distributed Stochastic Gradient Driven Distributed Active Noise Control System," *IEEE Open Journal of Signal Processing*, vol. 7, pp. 643–658, 2026. doi: 10.1109/OJSP.2026.3701314
My role: wrote the manuscript up to the simulation section (introduction, background, proposed algorithm and performance analysis) from the methodology provided by my supervisor, and made Figures 1 and 2.

Conference

[C1] **A. S. Shamkuwar** (first author), P. Bhardwaj, N. Kumar, and M. Singh, "Early Sepsis Detection from ICU Data Using Multimodal Deep Learning," in *2026 Int. Conf. on Intelligent Processing, Hardware, Electronics, and Radio Systems (CIPHER)*, Jalandhar, India, Feb. 2026, pp. 1–5. doi: 10.1109/CIPHER70417.2026.11523895
Presented at the conference.

[C2] A. Sharma, **A. S. Shamkuwar**, U. Anand, R. Kumar, N. Kumar, and M. Singh, "Spatio-Temporal Node Localization in Wireless Sensor Networks Using Hybrid Recurrent-Graph Neural Network," in *2026 CIPHER*, Jalandhar, India, Feb. 2026, pp. 1–5. doi: 10.1109/CIPHER70417.2026.11524038
My role: ran the simulations and produced the results (four graph and recurrent models across 15 mobility and noise settings).

RESEARCH EXPERIENCE

M.Tech. Thesis: MultimodalSepsisNet (early sepsis prediction, PhysioNet/CinC 2019) — Jun 2025 – May 2026
Supervisor: Dr. Manjeet Singh, NIT Jalandhar

- Designed a multimodal network that splits 40 clinical variables into four modalities (vital signs, laboratory values, missingness patterns, static demographics), with TCN–BiGRU, BiGRU and feedforward encoders fused by eight-head cross-modal attention (3.0M parameters).
- Built a leakage-free pipeline on 39,721 ICU records: patient-level stratified splits, imputation, normalisation and class-imbalance handling, evaluated at 6-hour and 12-hour prediction horizons.
- Achieved the highest AUC-ROC and AUPRC among eight baselines (LR, DT, RF, XGBoost, RNN, LSTM, GRU, TCN) at both horizons: 0.9535 and 0.7740 at 12 h (+0.087 AUPRC over XGBoost) and 0.9407 and 0.7441 at 6 h, with 95% bootstrap confidence intervals.
- Ablations showed that removing the laboratory stream, missingness stream or cross-modal attention lowers AUPRC by 0.224, 0.083 and 0.069; attention analysis showed focus on early ICU hours and admission laboratory results.

ACADEMIC APPOINTMENTS

Teaching Assistant – Deep Learning Laboratory, *NIT Jalandhar*

- Conducted hands-on PyTorch lab sessions. Experiments covered a multilayer perceptron with manual back-propagation, CNN image classification, transfer learning with pretrained networks, RNN/LSTM/GRU sequence models, autoencoders, and regularisation (dropout, batch normalisation, data augmentation).

- Designed assignments that asked students to compare architectures, run ablations, analyse training curves and report reproducible results.
- Held doubt-solving sessions.

Teaching Assistant – Machine Learning Laboratory, *NIT Jalandhar*

- Conducted weekly Python lab sessions (NumPy, pandas, scikit-learn). Experiments covered linear and logistic regression, k-NN, decision trees and random forests, SVMs, k-means clustering, PCA, and model evaluation with cross-validation, ROC curves and confusion matrices.
- Designed assignments around the bias–variance trade-off, regularisation, feature engineering, hyperparameter tuning and class imbalance, each ending in a short report on a real dataset.
- Held doubt-solving sessions.

PROJECTS

Robust Brain Tumor Segmentation with 3D Deep Learning (BraTS adult glioma MRI) Ongoing

- Building a reproducible 3D U-Net pipeline that segments tumor sub-regions (whole tumor, tumor core, enhancing tumor) from four MRI sequences: T1, T1c, T2 and FLAIR.
- Baseline design: brain-masked per-patient normalisation, patient-level splits, 3D patch-based training on a single GPU, and Dice and HD95 evaluation with error analysis.
- Next: measure how segmentation performance holds up when one or more MRI sequences are missing.

Human Activity Recognition: Signal Processing vs. Deep Learning (code) Python, PyTorch

- Engineered 63 time- and frequency-domain features (Welch PSD, zero-crossing rate, spectral entropy) from raw 9-channel accelerometer and gyroscope signals; compared Random Forest, 1D-CNN and bidirectional LSTM.
- Fixed a sitting/standing failure using feature-importance analysis and gravity-orientation features, raising accuracy from 89% to 95% (macro F1 0.94).
- Validated with leave-one-subject-out cross-validation over 30 subjects (93.9% $\pm$ 5.3% accuracy); deployed with FastAPI and Streamlit.

Real-Time Pothole Detection and Geotagging (B.Tech. final-year project; code) Python, OpenCV

- Built a real-time detector running a YOLO-tiny model through OpenCV DNN with CUDA FP16 inference, using confidence thresholding, non-maximum suppression and box-size filtering to suppress false detections.
- Saved annotated frames with location coordinates at fixed intervals and recorded result video with a live FPS overlay.

Finance RAG System over SEC EDGAR Filings Python, FastAPI, React

- Built retrieval-augmented question answering over 10-K/10-Q filings: automated ingestion and cleaning, tables kept as single chunks, hybrid BGE/FAISS and BM25 retrieval with reciprocal rank fusion, and citation-grounded Gemini answers.
- Served through a Dockerized FastAPI API and React interface; wrote routing and retrieval evaluation scripts.

RELEVANT COURSEWORK

Machine learning: Machine Learning, Applied Machine Learning, Deep Learning, Pattern Recognition
Signal and image processing: Statistical Signal Processing, Advanced Signal Processing, Digital Signal Processing, Digital Image Processing
Mathematical foundations: Probability and Stochastic Processes, Applied Linear Algebra and Optimization
Computing and systems: Data Structures and Algorithms, Object-Oriented Programming, Embedded Systems, Control Systems

TECHNICAL SKILLS

ML/DL: Machine Learning, Deep Learning, Image Processing, PyTorch, Scikit-learn, OpenCV
GenAI/LLM: LangChain, RAG, Prompt Engineering
Languages: Python, C, C++, MATLAB, SQL, HTML, CSS
Data and visualisation: NumPy, Pandas, Matplotlib, Seaborn, Plotly
Tools and deployment: FastAPI, Streamlit, Git, Docker

REFERENCES

Available on request.